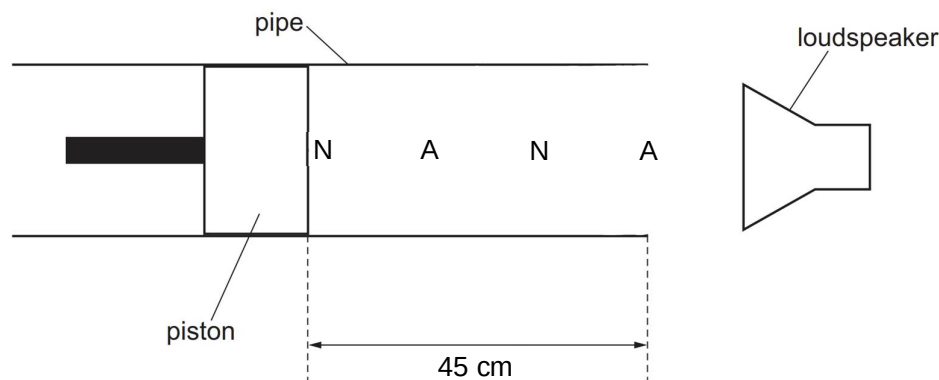


- 3 (a) (i) For maximum intensity, waves must be in phase at detection [1]
 Phase difference at source means waves are not in phase / have 90° phase difference at O. [1]
- (ii) Point B labelled on the line PQ below O and distance OB = distance OA [1]
- (iii) $\lambda = (3.0 \times 10^8) / (2.5 \times 10^{10}) = 0.012 \text{ m}$ [1]
 $x = \lambda D / a = (0.012)(2.3) / (0.18) = 0.153 \text{ m}$ [1]

(b) (i)



N at closed end and A at open end [1]
 Correct number of N and A, and distance between N and A is constant [1]

- (ii) Lowest frequency happens when $\frac{1}{4} \lambda = 0.45$ [M1]
 $\lambda = 1.8 \text{ m}$ [A1]
 $f = 330 / 1.8 = 183 \text{ Hz}$
- (iii) the node-antinode distance is longer OR wavelength is longer [M1]
 (speed of sound is constant so) the frequency is lower [A1]
- (iv) End correction [1]

(c) Path difference = $0.4 \text{ m} = 2.5 \lambda$. Therefore, P is a minima [1]

Intensity $\propto$ amplitude² and intensity $\propto (1/r)^2$, therefore amplitude $\propto 1/r$

$$\frac{A_y}{A_x} = \frac{r_x}{r_y} = \frac{1.4}{1.8} \quad [1]$$

$$A_y = \frac{7}{9} A_x$$

$$\frac{I_{\text{resultant}}}{I_x} = \left(\frac{A_x - A_y}{A_x} \right)^2$$

$$\frac{I_{\text{resultant}}}{4.5 \times 10^{-6}} = \left(\frac{2}{9} \right)^2 \quad [1]$$

$$\text{Resultant } I = 2.22 \times 10^{-7} \text{ W m}^{-2}$$

